The D6-D8 series limit and its Painleve III(D8) limit

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex (Painlevé-III Monodromy Maps Under the D6 to D8 Confluence and Applications to the Large-Parameter Asymptotics of Rational Solutions); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: All delivered mathematical text is the author own, copied verbatim from the pinned journal source. Statement source lines 471-477 with the author own proof at 478-481 (label thm:D_6-D_8-series). Printed number 2.1 (the census published value, independently reproduced by the source walk: counter 0 of section 2). The statement and proof spans touch neither the public statement [255,284] nor the public proof boxes [1767,2887], [3074,3310] of problem_id 902, and the spans carry no unstaged citation.

Editorial note. The label thm:D_6-D_8-series names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own Theorem environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, bibliography backend none; any literature citation carried by the delivered span is disclosed in the notes of this card.

Editorial note. Revision r002: re-assembled once the delivery chain admitted "$" to the external-label whitelist (Chief ruling §8.8.136, 2026-10-15 01:5x), which two pointer labels of this card need (eq:D_6-parameters, eq:PIII-D_8). Nothing else changed.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The sequence of solutions U_n. Supplies background that the delivered proof uses. The entry is declared as a dependency; it is not re-proved on the card, and the load-bearing steps of the counted statement are carried out inside the delivered span.

Source reference. Original equation 2: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 370–370. Navigation pointer invoked at line(s) 480: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original equation 1: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 131–131. Navigation pointer invoked at line(s) 475: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

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Source reference. Original equation 2: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 390–390. Navigation pointer invoked at line(s) 480: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

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lines 365–365. Navigation pointer invoked at line(s) 472: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original equation 2: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 385–385. Navigation pointer invoked at line(s) 479: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original equation 2: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 403–403. Navigation pointer invoked at line(s) 480: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original equation 2: operations/codex-takeover-01a12558/corpus-repair-20261010/research-line/harness-ready-20261011/sources/sigma-2024-019-painlev-233-iii-monodromy-maps-under-the-d-6/pinned-original-source.tex, source lines 418–418. Navigation pointer invoked at line(s) 479: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Theorem 2.1. *Fix $\alpha \in \mathbb{C}$ and $\beta \in \mathbb{C}$ and let $\{U_n(z;\alpha,\beta)\}_{n=1}^{\infty}$ be a sequence of solutions of* (2) *that are analytic at the origin $z = 0$ and suppose that*

$$\lim_{n\to\infty} U_n(0;\alpha,\beta) = \upsilon_{\infty,0} = \upsilon_{\infty,0}(\alpha,\beta) \neq 0.$$

*Then there exists a radius $\rho > 0$ such that for all n sufficiently large $U_n(z;\alpha,\beta)$ is analytic for $|z| < \rho$ and such that $U_n(z;\alpha,\beta) \to U_\infty(z;\alpha,\beta)$ as $n \to \infty$ uniformly for $|z| < \rho$, where $U(z) = U_\infty(z;\alpha,\beta)$ is the unique solution of the Painlevé-*III(D_8) *equation* (1) *that is analytic at the origin with $U_\infty(0;\alpha,\beta) = \upsilon_{\infty,0}$.*

Proof. Let $\{\upsilon_{n,k}\}_{k=0}^{\infty}$ denote the power series coefficients of $U_n(z;\alpha,\beta)$ as in (2). Define Υ_0 by $\Upsilon_0 > 2|\upsilon_{\infty,0}|$ (say), and obtain the subsequent coefficients $\{\Upsilon_k\}_{k=1}^{\infty}$ via (2)–(2). Comparing (2)–(2) then shows that for all n sufficiently large, $|\upsilon_{n,k}| \leq \Upsilon_k$ holds for all $k = 0, 1, 2, \ldots$. For each fixed $k = 0, 1, 2, \ldots$, the recurrence relations (2)–(2)

together with the limit $v_{n,0} \to v_{\infty,0}$ show that $v_{n,k}$ tends to a limiting value $v_{\infty,k}$ as $n \to \infty$, with $|v_{\infty,k}| \leq \Upsilon_k$. The convergence of $U_n(z;\alpha,\beta)$ to a limiting analytic function $U_\infty(z;\alpha,\beta)$ with $U_\infty(0;\alpha,\beta) = v_{\infty,0}(\alpha,\beta)$ then follows by dominated convergence. That the limiting analytic function $U_\infty(z;\alpha,\beta)$ is a solution of (1) follows from passing to the limit in each term of (2) using (2). That this solution is the unique analytic solution of (1) with the specified value at $z = 0$ then follows from passing to the limit in the recurrence relations (2)–(2). □